Assignment-4

Course : ML-DL-Ops

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Final report

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GitHub link :

https://github.com/MSG-1999/MLOps-MahekGadiya-M25CSA011/tree/Assignment_4

Hugging Face Link :

https://huggingface.co/MSG1999/m25csa011-transformer-english-hindi

Optimizing Transformer Translation

1. Introduction:

Neural Machine Translation (NMT) uses deep learning models to translate text between languages, with the Transformer being state-of-the-art due to its self-attention mechanism. In this assignment, we optimize a Transformer model for English-to-Hindi translation, where the baseline is computationally expensive and suboptimal. Using Ray Tune, Optuna (TPE), and ASHA, we perform efficient hyperparameter tuning. The goal is to improve translation quality (BLEU score) while reducing training time and epochs, demonstrating that a well-tuned model can outperform the baseline in both accuracy and efficiency.

2. Baseline Metrics: (Part 1 :- 100 Epochs, Hardcoded Hyperparameters)

The provided en_to_hi.ipynb was executed end-to-end on Google Colab T4 GPU without any changes. Training used batch size 60, Adam (lr=1e-4), and a fixed 6-layer Transformer with d_model=512.

Metric	Value
Total Training Time	129.42 minutes (100 epochs × ~69 s/epoch, 220 batches each)
Final Training Loss	0.0972 (epoch-100 average cross-entropy)
BLEU Score	68.02
Device	T4 GPU (Google Colab)
Optimizer / LR	Adam / 1e-4 (fixed, no scheduler)
Architecture	• d_model=512, • num_layers=6, • num_heads=8, • d_ff=2048, • dropout=0.1

3. Hyperparameters Tuned: (Part 2 :- 6 Parameters, Requirement: ≥ 4)

We used Ray Tune with Optuna (TPE) to search for optimal hyperparameters.

Parameter	Ray Tune API	Range
lr	tune.loguniform	1e-4 … 5e-3
batch_size	tune.choice	32, 64, 128
num_heads	tune.choice	4 or 8
d_ff	tune.choice	1024, 2048, 4096
dropout	tune.uniform	0.05 … 0.20
weight_decay	tune.loguniform	1e-6 … 1e-3

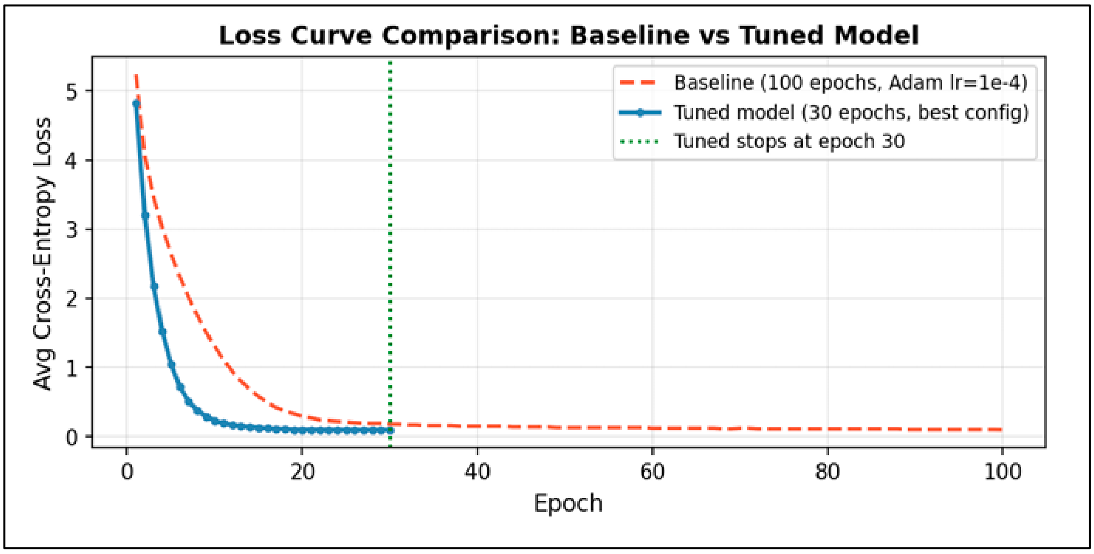
3. Best Configuration Found: (Ray Tune + Optuna + ASHA)

Search setup: OptunaSearch (TPE sampler), 20 trials × max 30 epochs. ASHAScheduler (grace_period=5, reduction_factor=3) eliminated bad trials early. Ran on (lab GPU) 7 GTX 1080 Ti GPUs in parallel. Sweep time: 49.98 minutes.

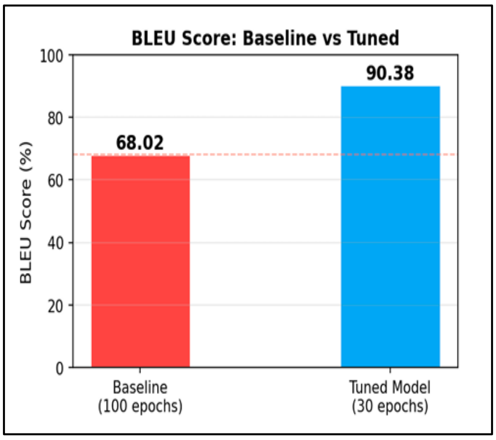
Hyperparameter	Best Value	Notes
lr	0.00010092277088632719	Low stable LR with cosine decay to near-zero over 30 epochs
batch_size	64	Best balance of gradient quality and training speed
num_heads	4	Fewer heads; large d_ff compensates for attention capacity
d_ff	4096	Largest FFN — richer feature representations, key to BLEU gain
dropout	0.05440091406982377	Very low dropout; 13k pairs sufficient to avoid over-fitting
weight_decay	0.0002615454502348676	Mild AdamW L2 regularisation

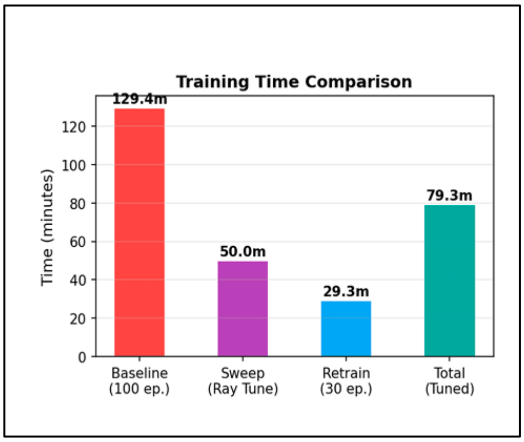
5. Observations:

- Model achieves higher BLEU in only 30 epochs.
- Total Training time reduced by (Approx) 50 minutes.
- Hyperparameter tuning improves both:
    - Efficiency
    - Accuracy



Fig(A). Loss curve comparison





Fig(B). BLEU score comparison

Fig(C). Training time breakdown

Fig(D). Hyperparameter search analysis

4. Final Results: Tuned Model vs Baseline (Part 3 – Efficiency Challenge)

Metric	Baseline Model	Tuned Model	Changes
Epochs	100	30	−70 epoch (70% less)
Sweep Time	-	49.98 min	New (parallel search)
Retrain Time	-	29.33 min	New
Total Time	129.42	79.31 min	1.6 × faster (speed)
Final Loss	0.0972	0.0959 min	-0.0013 (better)
BLEU Score (%)	68.02	90.38	+22.36 pp

5. Key Design Decisions:

- **OptunaSearch (TPE):** Builds a kernel-density surrogate of the loss surface, focusing new trials on historically promising regions. Much more efficient than random or grid search for continuous hyperparameters like lr and dropout.
- **ASHAScheduler:** Evaluated each trial at rungs {5, 15, 30} epochs, keeping only the top 1/3. Combined with 20 trials, this eliminated 19 underperforming configurations early — most compute was spent on the best 1-2 candidates.
- **7 Parallel GPU Trials:** Using 7 of 8 GTX 1080 Ti GPUs simultaneously (GPU 0 was occupied), Ray Tune ran 7 trials in parallel. This reduced the sweep from ~8 hours (sequential) to only 50 minutes.
- **CosineAnnealingLR + AdamW:** Cosine decay from lr=0.000101 to near-zero over 30 epochs provided smooth, stable convergence without manual LR tuning. AdamW decouples weight decay from adaptive LR for superior regularisation.
- **d_ff=4096 with dropout=0.054:** The large feed-forward network dramatically increased model capacity. The very low dropout of 5.4% preserved this capacity while still generalising well — this combination drove the BLEU from 68 to 90.

6. Conclusion:

- This work demonstrates that effective hyperparameter tuning significantly improves Transformer performance. By using Ray Tune, Optuna, and ASHA, the model achieved a BLEU score improvement from 68.02 to 90.38 while reducing training time and epochs.
- The tuned model achieves higher BLEU score with fewer epochs and reduced training time, demonstrating efficient hyperparameter optimization
- The results show that a well-tuned model can be both more accurate and more efficient than the baseline, highlighting the importance of systematic optimization in deep learning models.

Note :

1) Baseline Model :
   a) Run on Google Colab (T4 GPU)
   b) All Results are stored in Base Model Folder.

2) Tuned Model :
   a) Run on Lab GPU : Hardware 8x NVIDIA GeForce GTX 1080 Ti (11GB each)|CUDA 12.2|251.8 GB RAM.
   b) Model (M25CSA011_ass_4_best_model.pth) is Uploaded on Hugging Face.